The Summit Sabotage

Whodunnit? · Linear Equations & Slope · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Linear Equations & Slope	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Writing equations of lines: slope-intercept form, parallel and perpendicular lines.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Dr. Boone took it, in the Summit Hut, using this method: Dragged on a sled.

The reveal (read aloud): While everyone watched the race, **Dr. Boone** slipped into the **Summit Hut**, strapped the Golden Ski Pole to a **sled**, and dragged it down the back trail. Every alibi and method you cleared led to him. Slope-solid work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Coach Petra · Place: Ski Rental Shop
2	Method: Stuffed in a ski bag
3	Suspect: Mr. Lund · Place: Chairlift Station
4	Method: Carried on a snowmobile
5	Suspect: Ms. Ferrari · Place: Tubing Hill
6	Method: Hidden in a snowman
7	Suspect: Hank Moreau · Place: First Aid Cabin
8	Method: Tucked in a gondola seat
9	Suspect: Ms. Yamada · Place: Lodge Café
10	Method: Strapped to a snowboard

Answer Key

The Summit Sabotage · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Write the equation of the line through (0, 3) and (2, 7) in slope-intercept form.	**A.** $y = 2x + 3$	slope $= \frac{4}{2} = 2$ y-intercept = 3 $y = 2x + 3$	Coach Petra was fitting boots in the Ski Rental Shop. Six customers remember her.
2	Write the equation of the line with slope −2 that passes through (1, 4).	**C.** $y = -2x + 6$	$y - 4 = -2(x - 1)$ $y - 4 = -2x + 2$ $y = -2x + 6$	Every ski bag was checked at the lodge door. None held the Pole.
3	Write the equation of the line through (−2, 0) and (2, 2).	**A.** $y = \frac{1}{2}x + 1$	slope $= \frac{2}{4} = \frac{1}{2}$ use (2, 2): $2 = \frac{1}{2}(2) + b$, so $b = 1$ $y = \frac{1}{2}x + 1$	Mr. Lund was running the Chairlift Station. The lift log shows him at the controls.
4	Write the equation of the line parallel to $y = 3x - 4$ that passes through (0, 5).	**C.** $y = 3x + 5$	parallel lines have the same slope, 3 b = 5 $y = 3x + 5$	The snowmobile's cargo box is empty and covered in fresh snow. Nothing was carried.
5	Write the equation of the line perpendicular to $y = 2x + 1$ that passes through (4, −1).	**A.** $y = -\frac{1}{2}x + 1$	perpendicular slope $= -\frac{1}{2}$ $-1 = -\frac{1}{2}(4) + b$, so $b = 1$ $y = -\frac{1}{2}x + 1$	Ms. Ferrari was posing for photos on the Tubing Hill with a crowd of fans.
6	Write $4x - 2y = 6$ in slope-intercept form.	**D.** $y = 2x - 3$	$-2y = -4x + 6$ $y = 2x - 3$ $y = 2x - 3$	The snowman by the lodge was knocked over by kids. Only snow inside.
7	A table of values has the points (0, −1), (1, 2) and (2, 5). Write the equation of the line.	**B.** $y = 3x - 1$	slope $= \frac{3}{1} = 3$ (y goes up 3 each time) y-intercept = −1 $y = 3x - 1$	Hank Moreau was bandaging a skier's knee in the First Aid Cabin.
8	A plant is 4 inches tall and grows 2 inches each week. Write an equation for its height *y* after *x* weeks.	**C.** $y = 2x + 4$	start = 4, rate = 2 per week $y = 2x + 4$	The gondola seats were cleaned at 9 a.m. and nothing was left behind.
9	A line crosses the x-axis at (6, 0) and the y-axis at (0, 3). Write its equation.	**B.** $y = -\frac{1}{2}x + 3$	slope $= \frac{3 - 0}{0 - 6} = -\frac{1}{2}$ $y = -\frac{1}{2}x + 3$	Ms. Yamada was serving hot cocoa in the Lodge Café during the whole blackout.
10	Write the equation of the line through (−3, 2) and (5, 2).	**D.** $y = 2$	slope $= \frac{0}{8} = 0$ (horizontal line) $y = 2$	All the snowboards were checked at the rack. None had the Pole strapped to them.

Wrong-Answer Map

The Summit Sabotage · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	B. $y = 4x + 3$	used rise without dividing by run	Coach Petra was resting in the Summit Hut after a morning lesson. Her students waved from the window.
1	C. $y = 3x + 2$	swapped slope and intercept	Dr. Boone was warming up with tea in the Lodge Café. The barista remembers the extra honey.
1	D. $y = -2x + 3$	sign error in slope	Every sled was lined up and inspected. Nothing was dragged on any of them.
2	A. $y = 2x + 6$	sign error in slope	The sled tracks in the snow all lead to the kids' hill, so no sled hauled the Pole away.
2	B. $y = -2x + 2$	subtracted slope term instead of adding	Mr. Lund rode up to the Summit Hut to fix a heater. The repair ticket has his signature.
2	D. $y = -2x + 4$	used y-coordinate as intercept	Dr. Boone was reading the wind gauge at the Chairlift Station. The weather log has his notes.
3	B. $y = 2x + 1$	inverted slope (run over rise)	Dr. Boone was having his gloves checked in the First Aid Cabin. The visitor book has his name.
3	C. $y = \frac{1}{2}x + 2$	used wrong y-intercept	Ms. Ferrari was signing autographs at the Summit Hut. A line of fans stretched out the door.
3	D. $y = -\frac{1}{2}x + 1$	sign error in slope	The rescue sled has been locked in the shed since morning. It never carried the Pole.
4	A. $y = -\frac{1}{3}x + 5$	used perpendicular slope	Hank Moreau was restocking the first-aid kit in the Summit Hut. The checklist is initialed.
4	B. $y = 3x - 4$	copied intercept of given line	Dr. Boone was renting snowshoes at the Ski Rental Shop. The rental slip is stamped 9:06 a.m.
4	D. $y = -3x + 5$	sign error in slope	Fresh snow covers the sled path behind the lodge without a single mark.
5	B. $y = -\frac{1}{2}x - 1$	used y-coordinate as intercept	Ms. Yamada was dropping off hot chocolate at the Summit Hut. The guestbook shows her visit.
5	C. $y = -2x + 7$	used opposite slope, not reciprocal	The sled rental tags show every sled came back before the pistol was fired.
5	D. $y = \frac{1}{2}x - 3$	forgot the negative reciprocal sign	Dr. Boone was giving a snow forecast on the Tubing Hill to a line of kids.
6	A. $y = -2x + 3$	sign error in slope and intercept	A hiker's photo shows Ms. Ferrari sipping cocoa in the Summit Hut as the pistol sounded.
6	B. $y = 2x - 6$	did not divide intercept by -2	A ranger checked the sled rack and found only mittens.
6	C. $y = 2x + 3$	sign error	Photos show Dr. Boone at the Chairlift Station taking temperature readings right when the pistol fired.
7	A. $y = \frac{1}{3}x - 1$	inverted slope	Coach Petra was drawing a ski route on the chalkboard in the Summit Hut. Her drawing is still there.

7	C. $y = -3x - 1$	sign error in slope	A kid’s plastic sled was the only one out that day, and it carried just a thermos.
7	D. $y = 3x + 1$	sign error in intercept	Dr. Boone was drawing a storm map on a napkin in the Lodge Café. Three skiers watched him.
8	A. $y = 4x + 2$	swapped slope and intercept	Dr. Boone was sliding down the Tubing Hill to test the snow. A ranger timed the run.
8	B. $y = 2x$	left out starting height	Hank Moreau was warming up in the Summit Hut after a patrol run. His radio log says so.
8	D. $y = -2x + 4$	sign error in slope	The sled dogs stayed in the kennel all day, so no sled pulled anything off the mountain.
9	A. $y = -2x + 3$	inverted slope	Mr. Lund was eating lunch in the Summit Hut. Two skiers shared his table.
9	C. $y = -\frac{1}{2}x + 6$	used x-intercept as y-intercept	Sled runners leave deep grooves, and the snow near the exit is smooth.
9	D. $y = \frac{1}{2}x + 3$	sign error in slope	Dr. Boone was waxing a pair of skis in the Ski Rental Shop. The mechanic vouches for him.
10	A. $y = x + 2$	used slope 1 instead of 0	The sled by the lodge door was loaded with firewood only, and the crew unloaded it in front of everyone.
10	B. $y = 2x$	treated constant as slope	Ms. Yamada was hanging the new banner in the Summit Hut when the pistol went off.
10	C. $y = -2$	sign error	Dr. Boone was in the First Aid Cabin getting a bandage for a blister. The nurse’s log shows the time.